Supplementary Material

Higher methanotroph abundance and bottom-water methane in ponds with floating photovoltaic arrays

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Supplemental Methods

Water Column Microbial Cell Abundances

Water samples were preserved for flow cytometry by adding 1 μL of 25% glutaraldehyde to 1 mL of 20 μm pre-filtered pond water, shaking, incubating for 10 minutes in the dark at room temperature, and freezing at -80 °C. Fixed samples were diluted to 15-20X, stained with SYBR Green I, and run in triplicate on an Attune NxT Flow Cytometer. DNA-positive cell counts were analyzed using the *flowCore* (Ellis et al. 2025) and *ggcyto* (Van et al. 2018) packages in R (R Core Team 2014).

References

Ellis, B., P. Haaland, Hahne F, N. Le Meur, N. Gopalakrishnan, J. Spidlen, M. Jiang, and G. Finak. 2025. flowCore: flowCore: Basic structures for flow cytometry data. R package version 2.2.0.

Van, P., W. Jiang, R. Gottardo, and G. Finak. 2018. ggcyto: Next-generation open-source visualization software for cytometry. Bioinformatics **34**: 3951–3953.

R Core Team. 2014. R: A language and environment for statistical computing.

Table S1: Summary of model information for mixed models to describe temperature, dissolved oxygen, surface and bottom water dissolved methane (CH_4) concentration, methanotroph abundance, and methanogen abundance, and relative abundance of methanogens and methanotrophs in ponds with and without floating photovoltaics (FPV). All models include FPV presence or absence and day of year as fixed effects and the pond ID as a random effect. For each fixed effect, the estimated effect size is listed ± standard error of the effect, with the associated t-value below in parentheses.

Response	Observations (n)	Intercept	FPV Presence	Day of Year
Temperature	36	24.9 ± 2.06 (12.1)	3.91 ± 0.60 (6.46)	-0.03 ± 0.01 (-3.00)
Dissolved Oxygen	36	-19.2 ± 18.5 (-1.03)	32.2 ± 5.43 (5.92)	0.23 ± 0.08 (2.72)
Surface Water [CH_4]	36	5.61 ± 2.60 (2.16)	-0.82 ± 0.65 (-1.26)	-0.01 ± 0.01 (-0.62)
Bottom Water [CH_4]	36	631 ± 289 (2.18)	-373 ± 126 (-2.96)	-1.02 ± 1.33 (-0.77)
Surface water methanotroph abundance	24	-905833 ± 350459 (-2.59)	-370238 ± 105012 (-3.53)	6408 ± 1604 (4.00)
Surface water methanogen abundance	24	2584 ± 3146 (0.821)	-930 ± 943 (-0.987)	-3.49 ± 14.4 (-0.242)
Bottom water methanotroph abundance	24	-472732 ± 241959 (-1.95)	-298583 ± 72501 (-4.12)	4171 ± 1108 (3.77)
Bottom water methanogen abundance	24	16844 ± 22827 (0.738)	5980 ± 10592 (0.565)	-33.9 ± 101 (-0.335)
Sediment methanotroph abundance	23	0.120 ± 0.024 (4.96)	0.003 ± 0.008 (0.338)	0.000 ± 0.000 (-2.87)
Sediment methanogen abundance	23	0.246 ± 0.039 (6.26)	0.011 ± 0.024 (0.448)	0.000 ± 0.000 (-1.145)

Table S2: Summary of model information for mixed models to describe bottle incubations to determine rates of water column dissolved methane (CH_4) production or consumption in ponds with and without floating photovoltaics (FPV). All models include FPV presence or absence and if the bottle was light or dark as fixed effects and the pond ID as a random effect. For each fixed effect, the estimated effect size is listed ± standard error of the effect, with the associated t-value below in parentheses.

Response	Observations (n)	Intercept	FPV Presence/Absence	Light/Dark Bottle
Surface CH_4 Flux	12	-0.09 ± 0.07 (-1.27)	-0.08 ± 0.10 (-0.82)	0.08 ± 0.03 (3.01)
Bottom CH_4 Flux	12	-1.69 ± 4.14 (-0.41)	2.11 ± 5.83 (0.36)	-0.66 ± 0.88 (-0.75)

Table S3: Results of least-square means comparisons between ponds with and without floating photovoltaics. Models for which comparisons are made are described in Tables S1 and S2.

Comparison	Estimate ± SE	z-ratio	p-value
Temperature	-3.91 ± 0.60	-6.46	< 0.001
Dissolved Oxygen	-32.3 ± 5.43	-5.92	< 0.001
Surface Water [CH_4]	0.82 ± 0.65	1.26	0.208
Bottom Water [CH_4]	373 ± 126	2.96	0.003
Surface CH_4 Flux	0.08 ± 0.10	0.82	0.413
Bottom CH_4 Flux	-2.11 ± 5.83	-0.36	0.718
Surface water methanotroph abundance	370238 ± 105012	3.53	< 0.001
Surface water methanogen abundance	930 ± 943	0.987	0.324
Bottom water methanotroph abundance	298583 ± 72501	4.12	< 0.001
Bottom water methanogen abundance	-5980 ± 10592	-0.57	0.572
Sediment methanotroph abundance	-0.003 ± 0.008	-0.34	0.736
Sediment methanogen abundance	-0.011 ± 0.024	-0.45	0.654

Table S4: Results of Welch two-sample t-tests comparing diffusive methane fluxes and gas exchange velocities (k600) between the center and edges of ponds with and without floating photovoltaic arrays (FPV). We first compared pond locations across treatments, and then between locations and within treatments.

Comparisons Between Ponds with and Without FPV				
Measurement	Pond Location	df	t-value	p-value
Diffusive CH_4 Flux	Center	2.00	-1.03	0.413
Diffusive CH_4 Flux	Edge	2.00	-0.97	0.436
k600	Center	2.18	-0.75	0.525
k600	Edge	2.00	-1.03	0.412
Comparisons between pond Center and Edge within the same pond type				
Measurement	FPV Present/Absent	df	t-value	p-value
Diffusive CH_4 Flux	Present	2.28	-1.67	0.221
Diffusive CH_4 Flux	Absent	3.34	0.34	0.751
k600	Present	2.03	1.17	0.362
k600	Absent	3.42	-0.31	0.774

Table S5: Results of Welch two-sample t-test comparing potential rates of sediment CH_4 production between ponds with and without floating photovoltaic arrays.

Measurement	df	t-value	p-value
Sediment CH_4 Flux	2.75	-1.26	0.302

Table S6: Methanogen and Methanotroph Taxonomic Classification

Phylum	**Class**	**Order**
Halobacteriota	Methanosarcinia	Methanosarciniales_A_2632
Halobacteriota	Methanomicrobia	Methanomicrobiales
Methanobacteriota_A_1229	Methanobacteria	Methanobacteriales
Thermoplasmatota	Thermoplasmata_1773	Methanomassiliicoccales
Methanobacteriota_B	Thermococci	Methanofastidiosales
Halobacteriota	Methanosarcinia	Methanotrichales
Halobacteriota	Methanocellia	Methanocellales
Thermoproteota	Methanomethylicia	Methanomethylicales
Pseudomonadota	Gammaproteobacteria	Methylococcales
Verrucomicrobiota	Verrucomicrobiae	Methylacidiphilales
Methylomirabilota	Methylomirabilia	Methylomirabilales

Table S7: FPV, pond, and DOY structure sediment methanogen and methanotroph communities with distinct drivers and effect strengths. PERMANOVA and β-dispersion results for sediment methanogen and methanotroph communities. Bray-Curtis dissimilarities were calculated from rarefied relative abundances. FPV, pond, and day of year (DOY) were modeled as main effects and interactions. R^2 indicates the proportion of variance explained; pseudo-F and p-values assess compositional differences. β-dispersion statistics test group variability. Residual R^2 is reported for each CH_4 group. Figures S3A and S3B show corresponding ordinations.

Habitat Type	**CH_4 Group**	**Variable**	**PERMANOVA**				**®-Dispersion**		**Figure**
			R^2	**Pseudo-F**	**p-value**	**Residuals**	**F**	**p-value**	
Sediment	Methanogens	FPV	0.11833	10.4653	0.001		3.8732	0.062	S3A
Sediment	Methanogens	Pond	0.28150	6.2241	0.001		0.6726	0.639	S3A
Sediment	Methanogens	DOY	0.09004	7.9632	0.001	0.38443	0.8903	0.46	S3A
Sediment	Methanogens	FPV:DOY	0.02538	2.2446	0.038				S3A
Sediment	Methanogens	Pond:DOY	0.10032	2.2182	0.002				S3A
Sediment	Methanotrophs	FPV	0.10058	7.6933	0.001		0.146	0.712	S3B
Sediment	Methanotrophs	Pond	0.20436	3.9078	0.001		0.8364	0.54	S3B
Sediment	Methanotrophs	DOY	0.13122	10.0370	0.001	0.44451	4.7482	0.006	S3B
Sediment	Methanotrophs	FPV:DOY	0.02217	1.6955	NS				S3B
Sediment	Methanotrophs	Pond:DOY	0.09717	1.8581	0.015				S3B

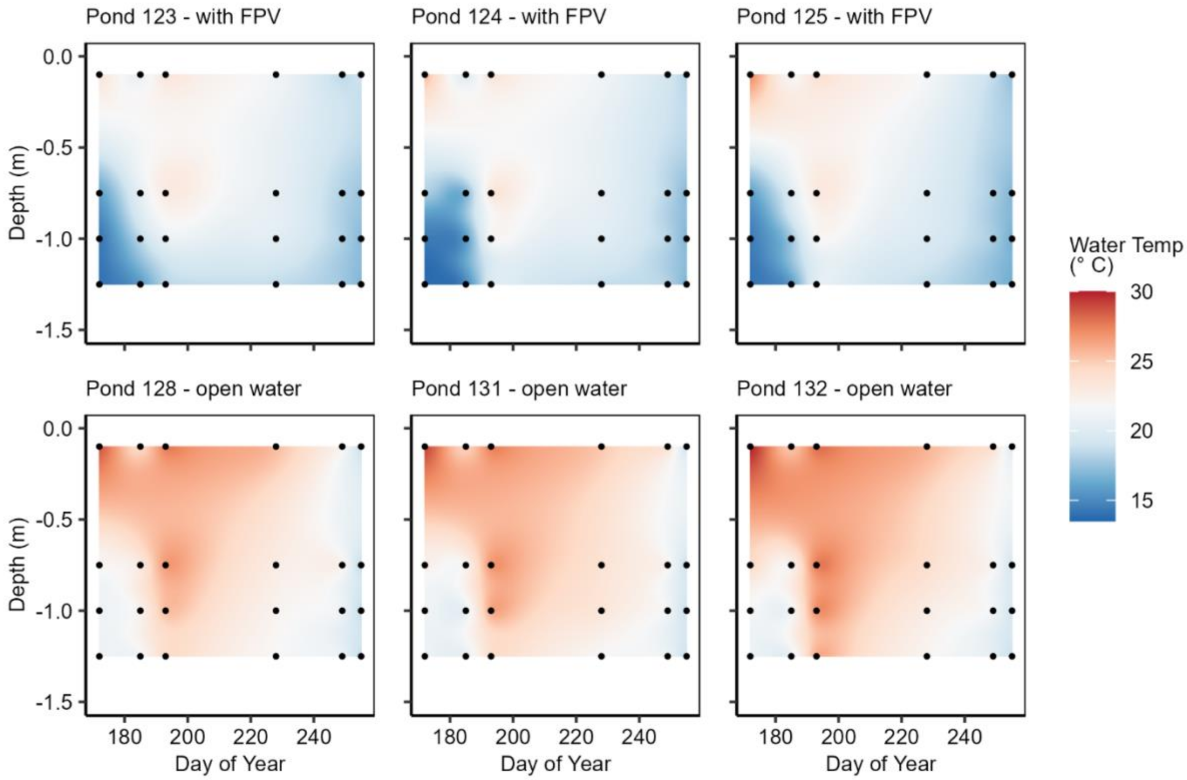

Fig. S1: Temperature profiles throughout summer 2024 in ponds with floating photovoltaic (FPV) arrays (top row) and ponds without FPV arrays (bottom row). Points indicate measurements and colors are interpolated based on those measurements.

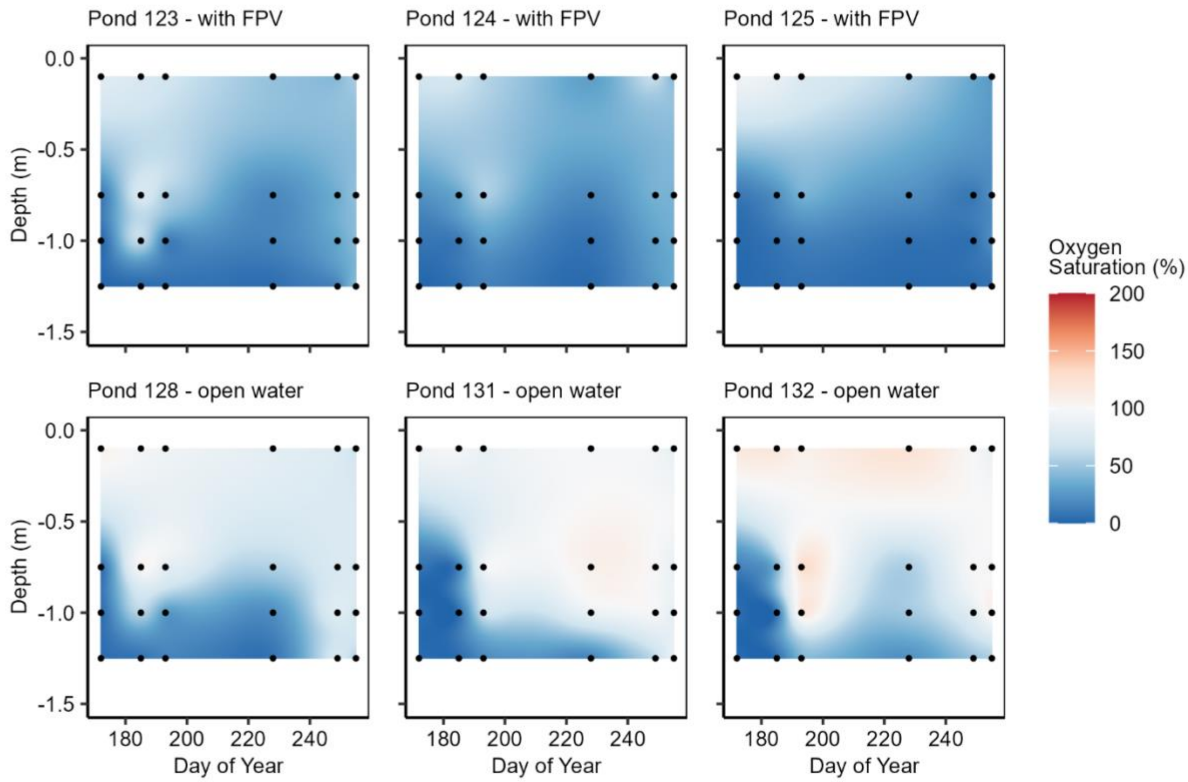

Fig. S2: Dissolved oxygen profiles throughout summer 2024 in ponds with floating photovoltaic (FPV) arrays (top row) and ponds without FPV arrays (bottom row). Points indicate measurements and colors are interpolated based on those measurements.

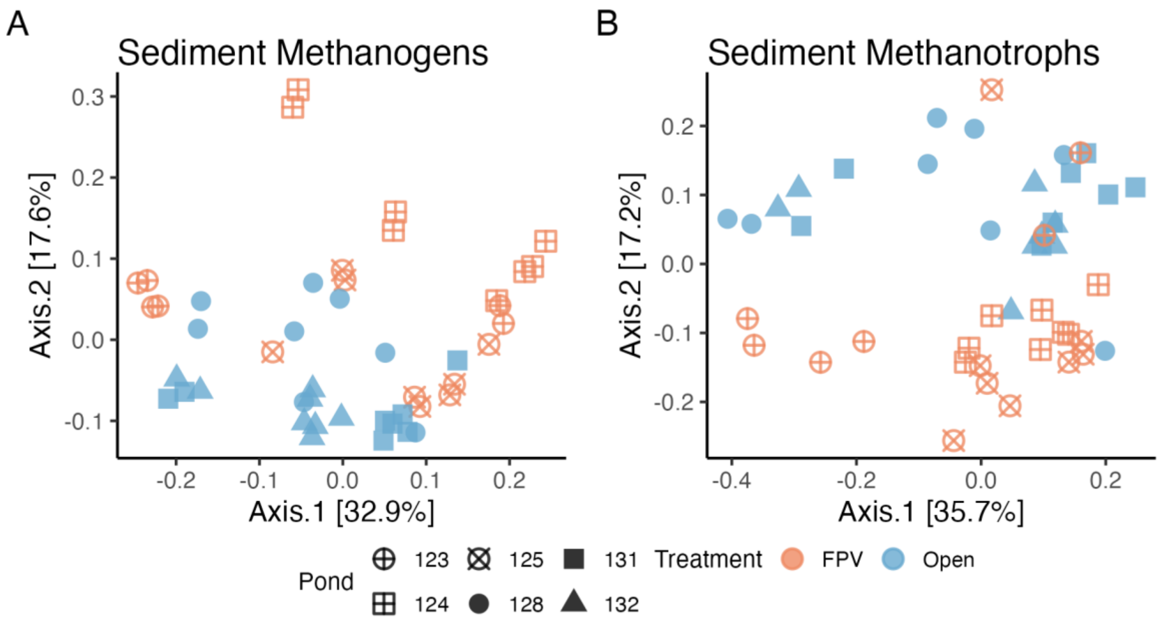

Fig. S3: Bray-Curtis PCoA of methane cycler communities within ponds. (A) Absolute abundance measures of water column methane cyclers were input and calculated with non-normalized absolute abundances of each ASV. (B) Sediments were rarefied to the minimum sequencing depth (20822 total sequence reads) and calculated relative abundance of each ASV. Sediments were collected in two replicates and are not merged.